

Bidurin Chakraborty

Principal Engineer | AI Product Owner

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EXPERIENCE SUMMARY

Strategic IT leader and architect with 14+ years of experience delivering high-scale cloud and Gen AI solutions for the automotive and software product sectors.

- **Leadership & Scale:** Manage P&L for ₹10Cr+ budgets and lead cross-functional teams of 8+ engineers; scaled product portfolios from 2 to 6 global enterprise offerings.
- **AI & Innovation:** Architected Semantic Search and RAG frameworks using Python and AWS Bedrock, automating workflows that reclaimed 160 person-days annually.
- **Cloud Transformation:** Delivered €2M+ in cost savings via large-scale data center exits and AWS/Azure migrations, reducing infrastructure OpEx by 30%.
- **Security & Compliance:** Expert in Zero Trust, IAM, and GDPR, ensuring secure document lifecycle management for 10,000+ global users across 12 manufacturing plants.
- **FOSS Strategy:** Championed Free and Open Source Software (FOSS) integration to reduce vendor lock-in and decrease development costs by 15%.

TECHNICAL ECOSYSTEM

AI/ML: RAG, Semantic Search, Prompt Engineering, LangChain, Transformers (Hugging Face), OpenAI API

Cloud & DevOps: AWS (Bedrock, S3, RDS, Lambda), Azure, GitHub Actions (CI/CD), Terraform (IaC)

Data & ECM: OpenText, Hyland, Saperion, Kofax, Snowflake, PostgreSQL, SQL Server, Power BI

Security: IAM, OIDC, TLS Encryption, Zero Trust, EU GDPR Compliance

PROFESSIONAL EXPERIENCE

MERCEDES-BENZ RESEARCH AND DEVELOPMENT INDIA (MBRDI)

Principal Engineer

Oct 2021 – Present

- **GenAI Strategy:** Conceptualized and deployed a Semantic Search engine using air-gapped local AI models, transforming legacy archival data into actionable business intelligence.
- **Digital Intelligence and Transformation:** Implemented OpenText and Kofax products to improve business processes through Document Lifecycle Management and automated workflows, increasing efficiency by 140%.
- **Business Process Automation:** Built AI-enabled validation tools that reduced manual document processing time by 160 person-days/year.
- **Infrastructure Lead:** Directed a global Data Center Migration for Daimler business units, achieving €2M+ savings through optimized infrastructure deployment.

Senior Consultant

Jun 2017 – Oct 2021

- **Solution Architecture:** Architected end-to-end Document Lifecycle systems (Imaging, Classification, Archiving) for global plants and functional areas including Sales, Aftersales, HR, and Logistics.
- **IT Governance:** Managed IAM and IT compliance (JDK, TLS, OIDC) for high-availability enterprise services.

DELL EMC

Senior Technical Support Engineer

Nov 2015 – Jul 2017

- **Technical Product Ownership:** Provided expert-level support and architectural troubleshooting for Documentum, Captiva, and InfoArchive ecosystems.
- **Performance Optimization:** Led complex software upgrades and database migrations, maintaining consistently high CSAT scores through proactive resolution.

ACCENTURE

Software Engineer

Jul 2012 – Nov 2015

- **Enterprise Development:** Engineered automated document management solutions utilizing Saperion, Kofax, Anydoc, and Captiva.
- **System Integration:** Developed and maintained backend services for Content Manager and Document Manager, ensuring seamless data flow across enterprise repositories.
- **Lifecycle Management:** Supported the full software development lifecycle (SDLC) for high-volume imaging and archival applications in the banking and manufacturing sectors.

PROJECT SPOTLIGHT: FOSS-BASED RAG ARCHITECTURE

Objective: Replace high-cost proprietary search with a secure, semantic AI retrieval system for automotive engineering documentation.

- **The Stack (FOSS-focused):** Orchestration in Python with LangChain.
 - LLMs: Integrated OpenAI API for cloud-native tasks and Mistral/Llama via Ollama for air-gapped, local inference.
 - Vector Database: ChromaDB and pgvector (PostgreSQL) for efficient embedding storage.
 - Embeddings: Hugging Face Sentence-Transformers (all-MiniLM-L6-v2) for cost-effective semantic mapping.
- **Architectural Impact:** Enabled "Chat-with-Documents" for legacy OpenText and Hyland repositories.
 - Eliminated expensive per-user licensing fees by utilizing a FOSS-centric deployment model.
 - Achieved a 40% improvement in document discovery speed compared to traditional keyword-based indexing.

EDUCATION & CERTIFICATIONS

- Bachelor of Technology (B.Tech), Electronics & Instrumentation — Techno India College of Technology
- Certified Scrum Master (CSM)
- AWS Certified — AI Practitioner
- SAFe 6.0 Scrum Certified